



Costume Design Guild

Fabric to Digital

Recreating real cloth behaviour in
Marvelous Designer

Block D

Week 5

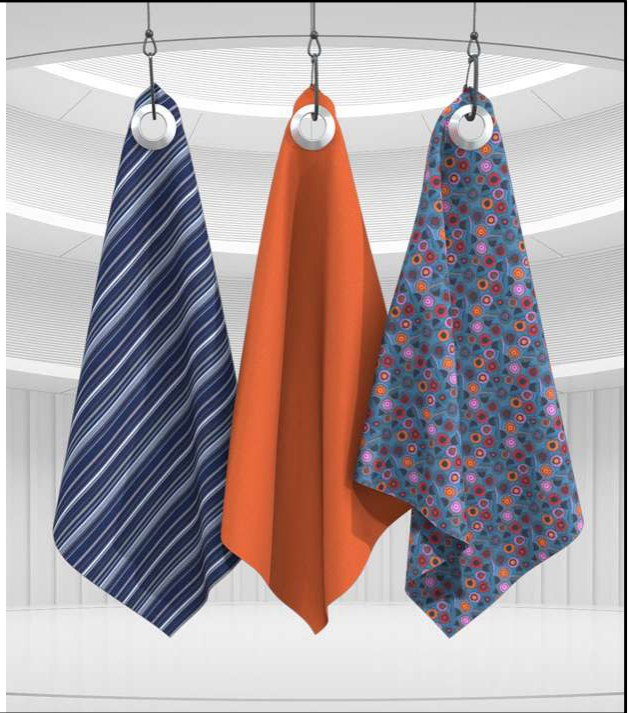
Download Marvelous Designer

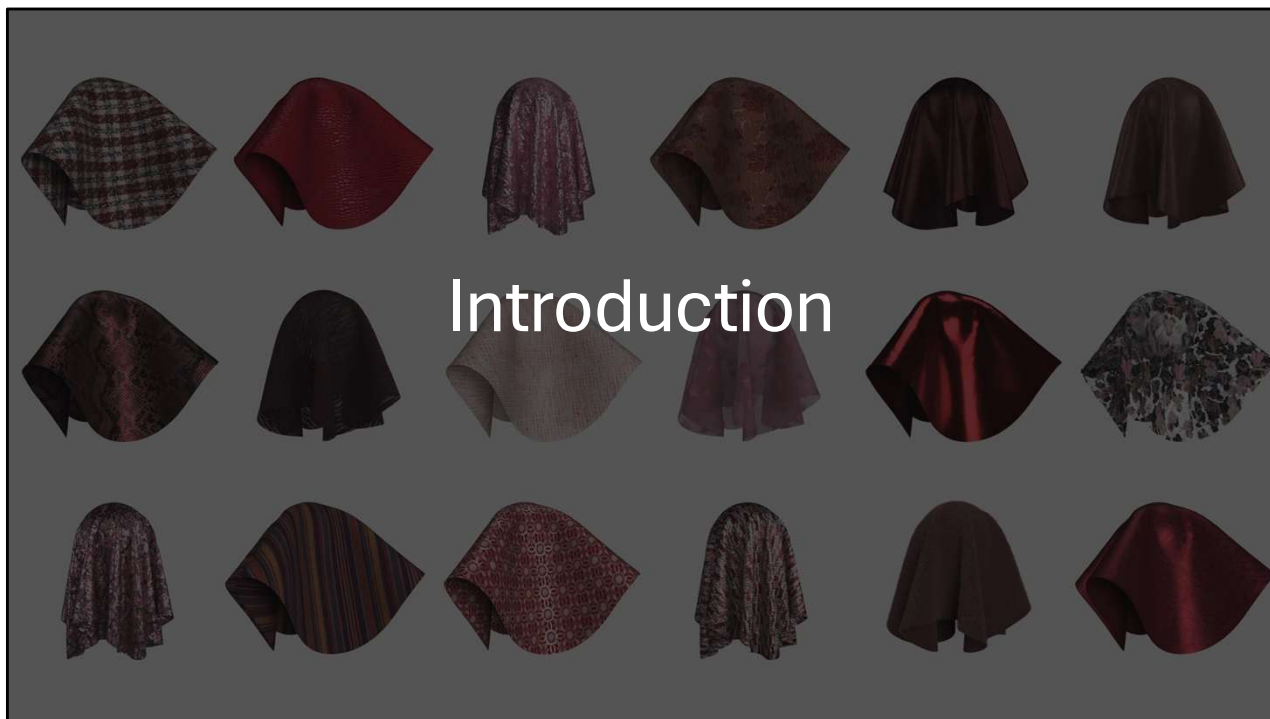


- Go to Marvelous Designer's website
 - Click account
 - Ask me to log in for you
 - Download
 - Open and log in again
 - Turn off your internet
- *Go to Teams and download the materials for this session*

Agenda

1. Introduction
2. Slider recap
3. Exploration
4. Closing and wrap up





The fabric

What do you notice?

- How heavy does it feel?
- How does it fall when you hold one end?
- What happens when you scrunch it?
- Does it spring back or just... stay?



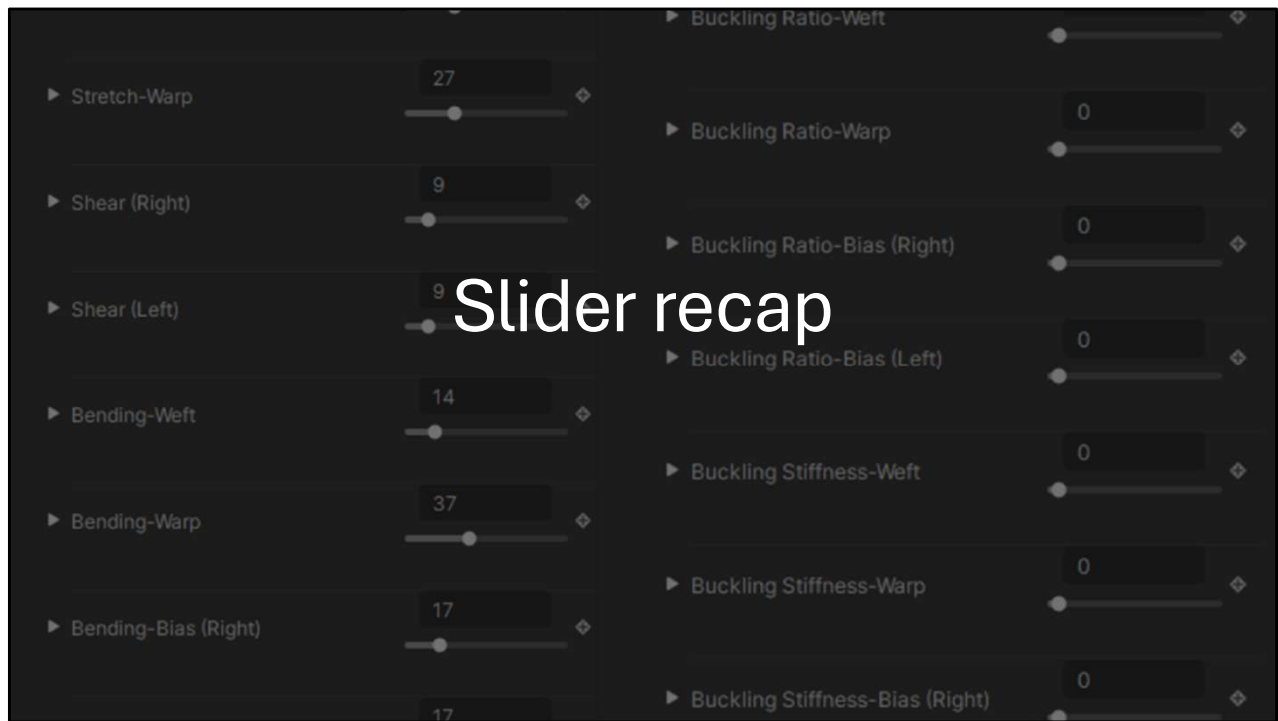
Today

You have:

+



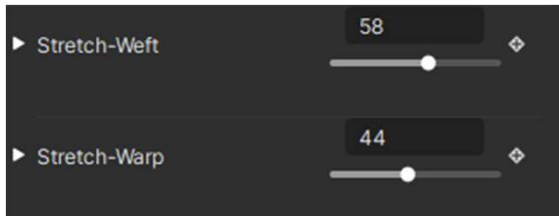
Make what's in MD match the fabric



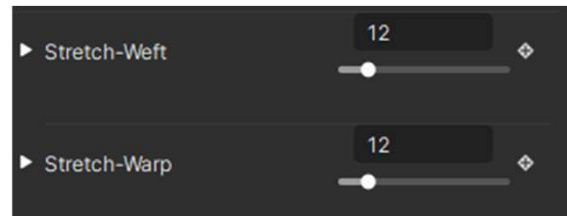
STRETCH

Resistance to being pulled

Wool



Knit Jersey



Higher number = fights being stretched.
Lower number = gives way easily.

Stretch is about the resistance to being pulled

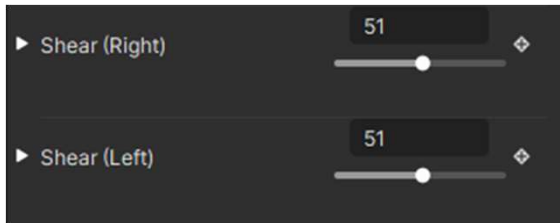
A higher number means it fights being stretched, so it doesn't stretch

A lower number means it gives way easily.

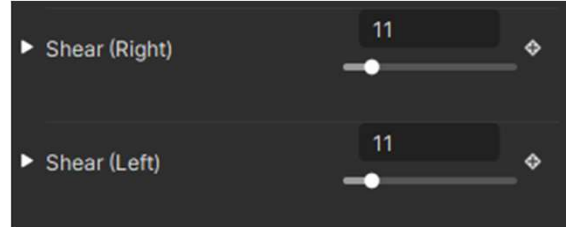
SHEAR

Resistance to diagonal distortion

Wool



Knit Jersey



Higher number = holds its grid shape.
Lower number = the weave shifts diagonally.

Then shear is about resistance to diagonal distortion

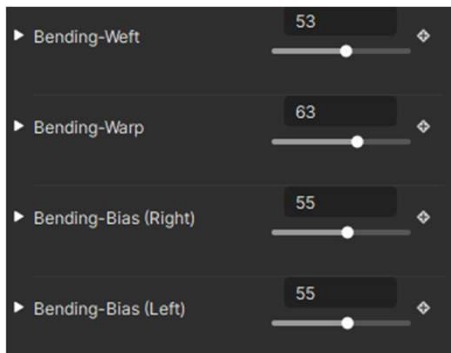
A higher number means it holds its shape

A lower number means it shifts and stretches diagonally

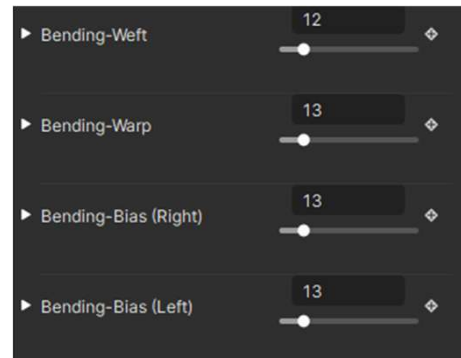
BENDING

Resistance to folding

Wool



Knit Jersey



Higher number = resists creasing, holds shape.

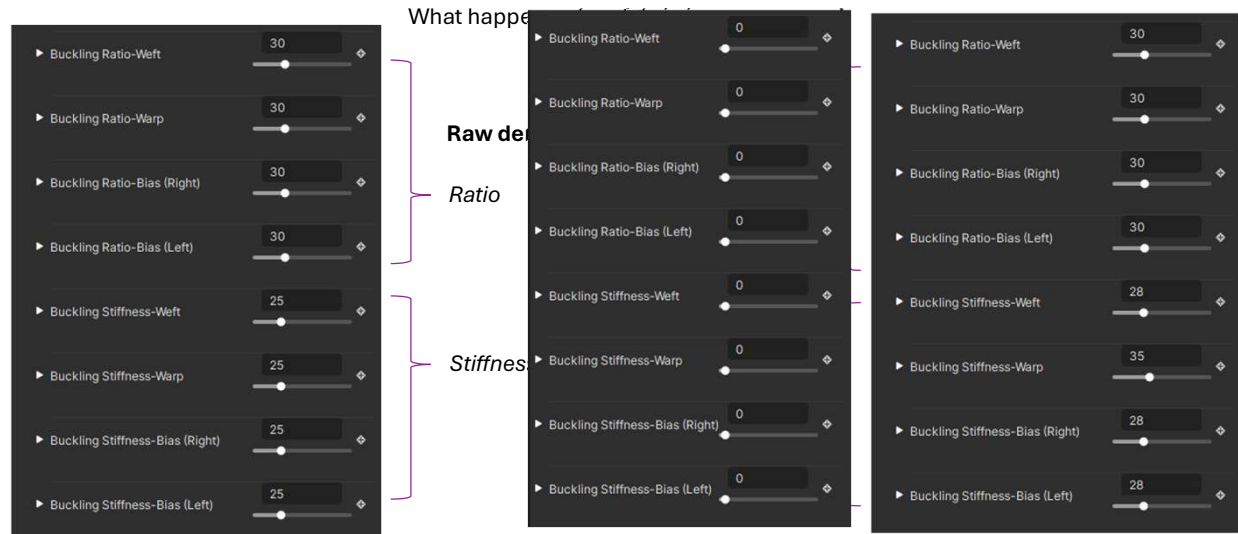
Lower number = collapses into folds easily.

Then bending is about resistance to folding

A higher number means it holds its shape

A lower number means it collapses into folds easily

BUCKLING



Buckling ratio = how much it resists before buckling.
Buckling stiffness = how sharp or soft the buckle is

11

Then buckling is about how much a fabric resists before buckling, buckling means creating creases

There's two types of buckling, ratio is about the resistance

And stiffness is about the sharp or softness of these buckles/creases

Hold the denim vertically and push the bottom upward with your other hand. The fabric resists briefly then buckles into a sharp angular fold — and stays there.

Say: "When denim is compressed — like at the back of a knee when sitting, or fabric bunching at a waist — it creates sharp, angular buckles that hold their shape. High stiffness means the buckle itself is crisp."

Do the same compression with the knit. It buckles immediately and softly — rounded, gentle folds that partly release when you stop pushing.

Say: "Same buckling ratio, but the knit's low bending resistance means the buckle is soft and rounded rather than sharp."

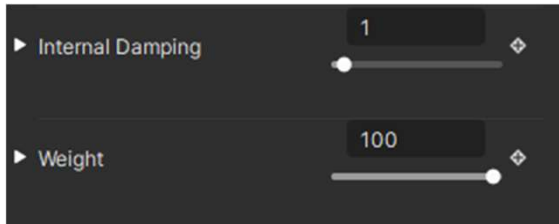
Notice the knit's warp stiffness is actually higher than denim here — that's what stops it from completely collapsing.

It buckles but bounces back a little."

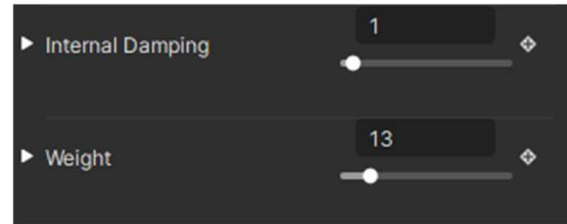
WEIGHT AND INTERNAL DAMPING

Gravity and *momentum*

Wool



Knit Jersey



Weight = how heavy the fabric is..

Internal damping = how quickly movement stops.

Low damping = keeps oscillating. High damping = settles immediately.

And finally, weight and internal damping.

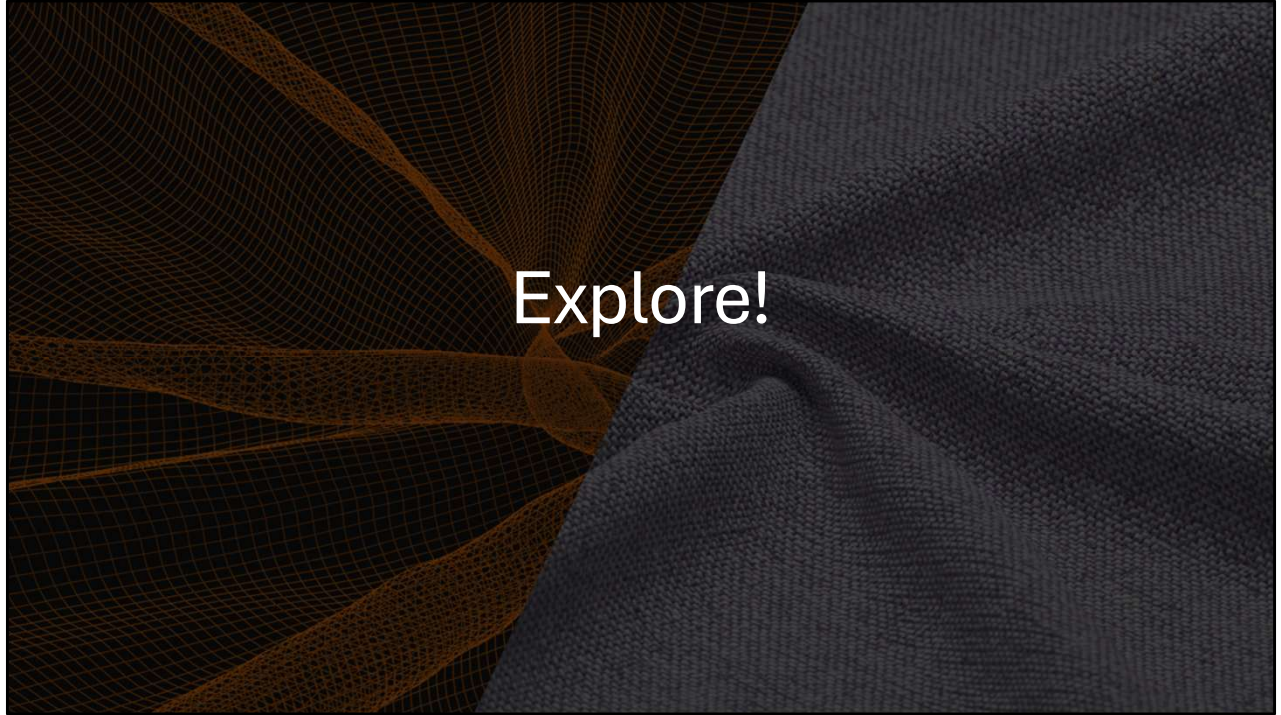
These are about gravity and momentum. These together is what makes fabric feel alive

Weight is about how heavy the fabric is, which is obvious. A denim is quite heavy. Heavier fabric falls faster, pulls folds downward more forcefully.

And a knit can be quite light

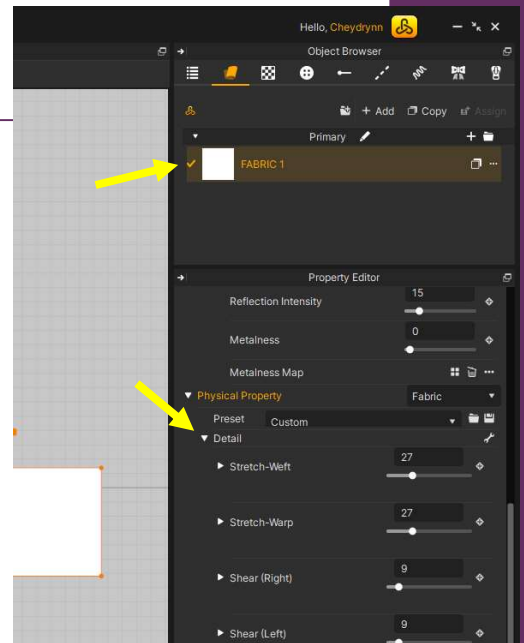
I believe the person who finetuned these sliders used weight to tune its settling behaviour so the internal damping isn't turned on

Internal damping means bounciness, because the knit is very light it bounces a lot (show purple metal sample)



Open MD

- Load in the session material file
- Click on the fabric
- Scroll for the sliders
- Press space for simulation



The fabric



Begin with bending and weight

The fabric



Begin with bending and weight



Reflection / wrap-up



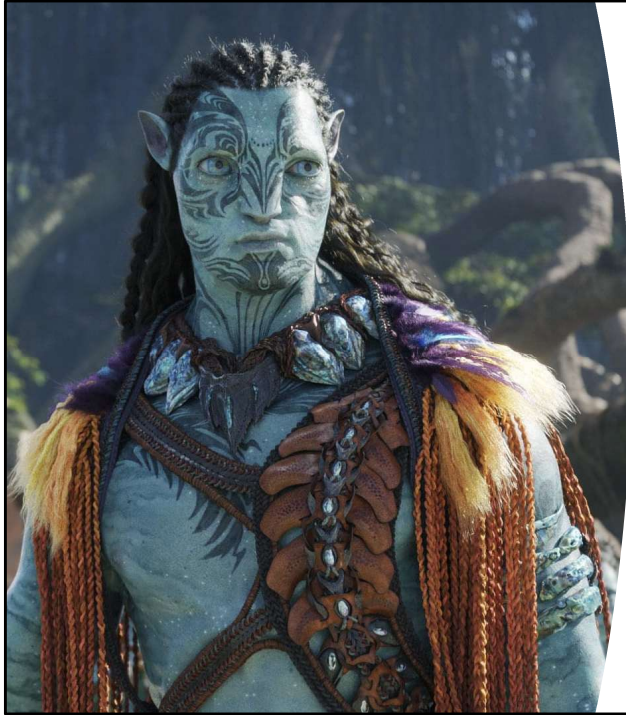
Key takeaways from today

- Every fabric has a fingerprint in these sliders
- Weight and damping control how fabric feels alive
- Bending is what separates structured from fluid
- Physical reference next to your screen beats guessing every time



Reflection prompt

- What settings got you closest?
- What surprised you?
- What would you change if you had more time?



Next guild:

Speculative materials for
extreme environments